

TEACHING NOTE

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KEL320

Air France Internet Marketing: Optimizing Google, Yahoo!, MSN, and Kayak Sponsored Search

Case Synopsis

Rob Griffin, senior vice president and U.S. director of search for communications consulting firm Media Contacts, was faced with the task of improving return on advertising dollars spent for Air France. Air France had been running several search engine marketing (SEM) campaigns to increase its share in the U.S. air travel market. Griffin had access to detailed data about the current campaigns that he needed to analyze before he could recommend how to allocate ad dollars across the various search engines, and select appropriate keywords and bid strategies for placement.

Students will use actual data to develop their campaign recommendations, which will include calculations of key performance indicators, including cost per click and probability to produce a sale.

Learning Objectives

After analyzing and discussing the case, students should be able to:

- Analyze and manipulate data using pivot tables in Excel
- Determine strategies for obtaining maximum total online bookings through the channels available
- Use a portfolio application model to determine an optimal publisher strategy and copy improvement analysis

Position of Case in Course or Module Topical Sequence

The case is appropriate for graduate and executive students. It can be used effectively in online marketing courses or online marketing modules within marketing courses.

Excel pivot tables are an integral part of the case analysis, so students must either be able to use pivot tables prior to analyzing the case or time must be devoted to instructing students on pivot tables.

Supplementary Materials

FOR INSTRUCTORS

The following materials have been provided with this teaching note to aid instructors in leading this case:

- *Air France Solutions* (Excel)
- *Air France Slides* (PowerPoint)

FOR STUDENTS

The following materials for students are available with the case:

- *Air France Exhibits* (Excel)

The following may be used as supplemental material to accompany the case:

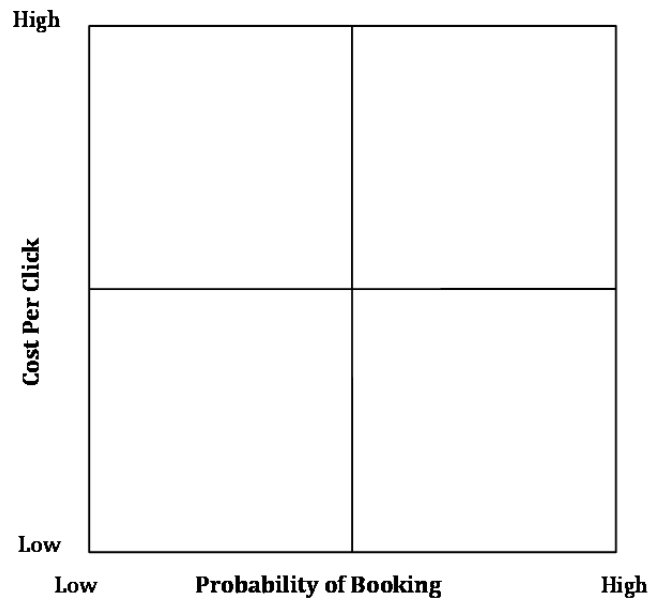
- Mark Jeffery, *Data-Driven Marketing: The 15 Metrics Everyone in Marketing Should Know* (Hoboken, NJ: John Wiley & Sons, 2010), chapter 7

Discussion Questions/Student Assignment

Use a pivot table and the accompanying Excel data file to compare the key performance metrics for each publisher (or search engine). Specific questions to consider include the following:

- Which publishers produce the best marketing campaign results?
- Which publishers should receive more marketing dollars? Which should receive less? Should any be eliminated?
- Which publishers would benefit from changing tactics, i.e., changing keyword, match type, and bid strategy?
- Which publishers would benefit from improved copy, i.e., improving the wording in the search ad or the content on the landing page?
- Kayak click data is not available through DoubleClick. How does Kayak compare to the other search engines? Would you recommend more or less funding for the Kayak SEM campaign?

You may find it useful to plot the publishers' results on the following 2×2 matrix with the dimensions of cost per click and probability of booking:



Use pivot tables and the accompanying Excel file to analyze the marketing campaigns in more detail. Specifically:

- Pick one publisher and identify which campaigns are working and which are not.
- Which keywords work best? Why? Which do not work?
- For Google-US, which match type and bid strategy work best? Why?
- What recommendations would you make to improve overall performance of specific campaigns?
- What are the specific effects of your recommended changes?

Pivot Tables

Pivot tables are essential to analyzing the case, which contains real SEM data. Basic pivot table explanation is provided in Exhibit 7 of the case, but many pivot table tutorials are available online from Microsoft and other companies for self-study or classroom use. Search for “Excel pivot table tutorial” and filter for the version of Excel that will be used in the classroom.

Caution: Do NOT calculate averages using the pivot table function “Summarize values by average”; it will not give the correct figure. This is because it does not weight the averages properly. The accurate average must be calculated by summing the numerator and the denominator and then dividing. The difference is illustrated in **Figure TN1**.

Figure TN1: Calculated Average vs. Pivot Table Average

Publisher Name	Pivot Table	Sum of Click Charges	Sum of Clicks	Calculated
	Average of Avg. Cost per Click			Sum of Click Charges / Sum of Clicks
Google-Global	\$2.22	\$120,946.71	72,895	\$1.66
Google-US	\$2.38	\$353,640.60	192,109	\$1.84
MSN-Global	\$2.15	\$12,160.36	11,217	\$1.08
MSN-US	\$2.87	\$16,098.49	10,808	\$1.49
Overture-Global	\$0.80	\$64,295.86	60,899	\$1.06
Overture-US	\$0.76	\$141,976.07	119,323	\$1.19
Yahoo-US	\$2.00	\$46,197.82	45,598	\$1.01
<i>Grand total</i>	<i>\$1.89</i>	<i>\$755,315.92</i>	<i>512,849</i>	<i>\$1.47</i>

Calculated averages will be used throughout the teaching note.

Excel Data

The instructor should ensure that students understand the data fields in the Excel spreadsheet.

The DoubleClick tab contains unaggregated data from multiple publishers. Relevant columns are labeled in **Figures TN2** through **TN5**.

Figure TN2: Excel Spreadsheet, DoubleClick Tab (View 1)

	Name of Publisher		Keywords		Name of Campaign	
	ID Unique to Publisher			Match Type		
1	Publisher ID	Publisher Name	Keyword ID	Keyword	Match Type	Campaign
2	K2615	Yahoo - US	43000000039657988	fly to florence	Advanced	Western Europe Destinations
3	K2615	Yahoo - US	43000000039651113	low international airfare	Advanced	Geo Targeted DC
4	K2003	MSN - Global	43000000019452431	air discount france ticket	Broad	Air France Brand & French Destinations

Figure TN3: Excel Spreadsheet, DoubleClick Tab (View 2)

	G	H	I	J	K	L
1	Keyword Group	Category	Bid Strategy	Keyword Type	Status	Search Engine Bid
2	Florence	uncategorized		Unassigned	Live	\$6.25
3	Low International DC	uncategorized		Unassigned	Paused	\$6.25
4	France	uncategorized	Position 2-5 Bid Strategy	Unassigned	Deactivated	\$0.00

Figure TN4: Excel Spreadsheet, DoubleClick Tab (View 3)

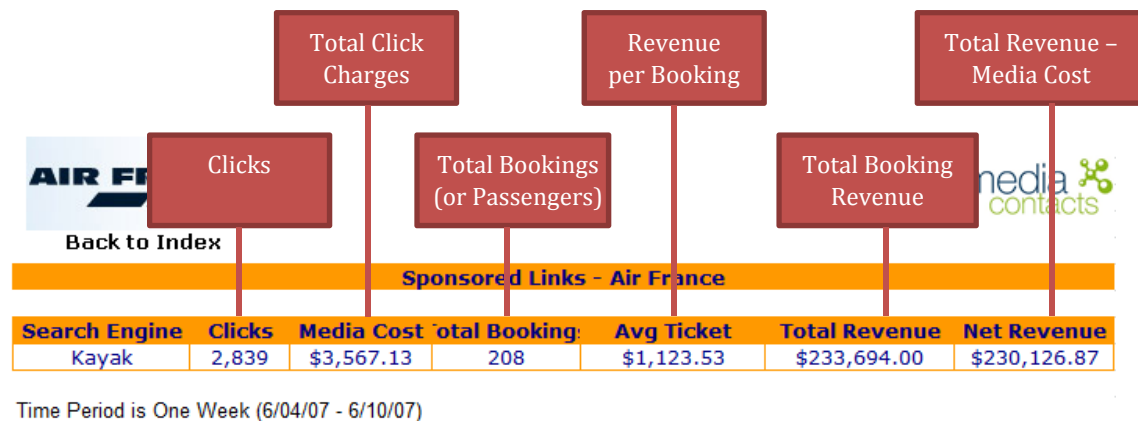
	M	N	O	P	Q	R	S
1	Clicks	Click Charges	Avg. Cost per Click	Impressions	Engine Click Thru %	Avg. Pos.	Trans. Conv. %
2	1	\$2.31	\$2.31	11	9.09%	1.27	900.00%
3	1	\$0.63	\$0.63	6	16.67%	1.00	100.00%
4	1	\$0.39	\$0.39	9	11.11%	1.11	100.00%

Figure TN5: Excel Spreadsheet, DoubleClick Tab (View 4)

	T	U	V	W
1	Total Cost/ Trans. Amount	Total Cost	Total Volume of Bookings	
2	\$0.26	\$8,777.95	\$2.31	9
3	\$0.63	\$1,574.20	\$0.63	1
4	\$0.39	\$390.15	\$0.39	1

The Kayak tab in the Excel file contains only aggregated data, as described in the case. In addition to understanding the various fields (see **Figure TN6**), students should understand the differences between Kayak and the other search engines. As described in the case, Kayak was a metasearch engine, which meant that it searched airline schedule and fare databases on airline sites, such as United.com, and directed users who wanted to make bookings to the airline site where they could complete their transactions. Advertising purchased keywords on Kayak, but ads could also be geotargeted, which enabled advertisers to determine the geographic location of website visitors and deliver different ads based on their location.

Figure TN6: Excel Spreadsheet, Kayak Tab



Case Analysis

SEARCH ENGINE MARKETING

Search engines such as Google, MSN, Overture, and Yahoo displayed ads next to the results of user searches. Advertisers purchased keywords related to their products; when a search query included those keywords, their ads were displayed. Typically advertisers paid a fee to the publisher (or search engine) every time someone clicked on their ad.

Google developed two key innovations in search. First, it developed a relevancy ranking to prioritize natural search (also called organic search) results. The relevancy ranking of a web page was determined by the number of other web pages linking to that page; in essence, at the top of its search results Google presented pages that not only matched the search criteria but also were the most relevant (or popular) as measured by links from other pages.

Google's second innovation was similar but related to advertising: in order to optimize its advertising revenue, it gave preferred placement to ads that attracted the most response. Like relevancy rates for search results, click-through rates prioritized the ads that were most popular with Google users. In order to discourage deceptive ads, Google adjusted the click-through rate to reflect the percentage of users who clicked on an ad and then left the target site within thirty seconds (called the bounce rate).

These two innovations helped Google gain popularity not only with users but also with advertisers that provided the revenue to support the free search service.

DEFINITIONS

Prior to doing the analysis it is essential that students understand the meaning of some key terms. **Table TN1** provides a glossary of five basic terms used in the case analysis and discussion.

Table TN1: Glossary

Term	Definition
Search engine marketing (SEM)	Purchasing keyword-sensitive sponsored links on search results pages of Google and other search engines. The links will only appear on results pages for searches that match purchased keywords.
Impressions	The number of times a sponsored link is displayed on search results pages.
Keywords	Words typed in the search engine by a user.
Match type	The order of keywords that will determine when an SEM ad appears. For an exact (or standard) match, the ad will appear only when the Internet user types the words in the exact order specified; for a broad (or advanced) match keywords can be typed in any order. Exact matches are less expensive than broad matches.
Bid strategy	An advertiser request for the placement of the sponsored link on the search results web page. As shown in Figure TN7 , a bid strategy of 1 to 3 targets the link in the first three listings on the page, and a 4 to 6 strategy places the link in the fourth, fifth, or sixth positions on a page. The higher the desired placement, the higher the cost. Positions 1, 2, and 3 are the most desirable because eye tracking studies show that the upper left corner of the page is viewed more frequently than any other part of the page (see Figure TN8).

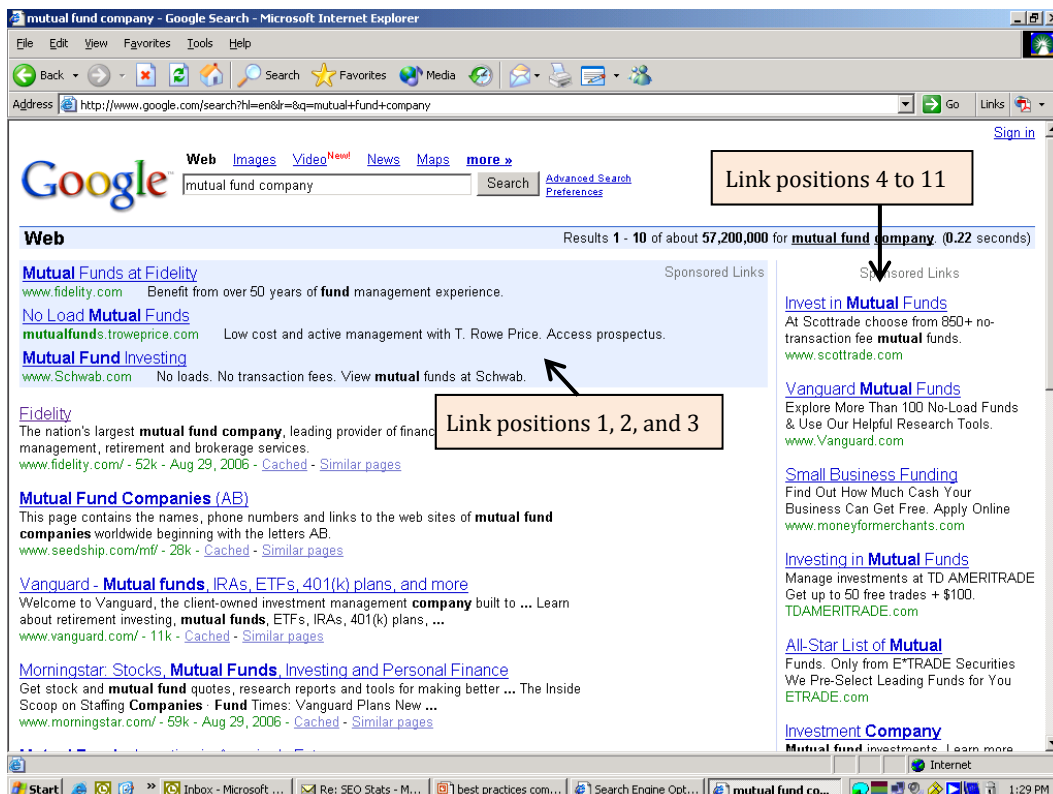
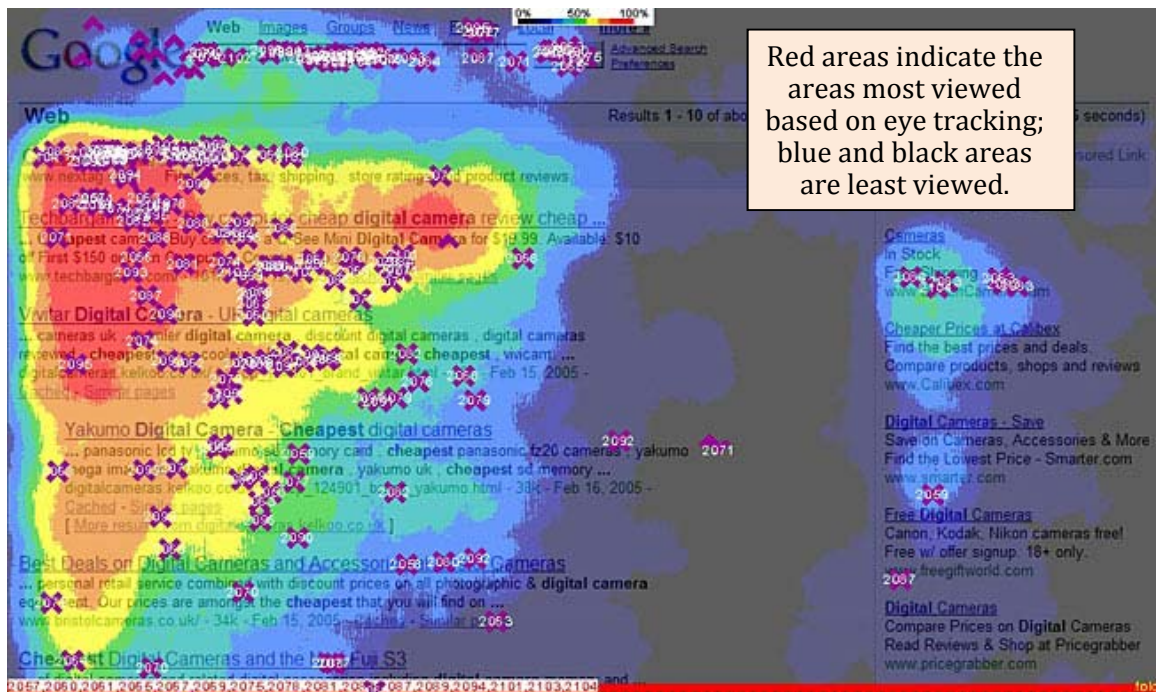
Figure TN7: Sample Google Search Results Web Page

Figure TN8: Heat Map of Google Search Results Web Page

Source: Adapted from the Enquiro Eye Tracking Reports, <http://www.enquiro.com>.

Seven key metrics are provided or calculated in the case (see **Table TN2**). The instructor should confirm that students understand not only how to calculate them but also what they mean and how to use them to make business decisions about optimizing SEM campaigns.

Table TN2: Explanation of Metrics

Metric	Calculation	Can be used to . . .
Cost per click (CPC)	Number of clicks / Cost of clicks	Compare costs differences among publishers
Click-through rate (CTR)	Number of clicks / Impressions	Determine the efficiency of sponsored link at generating clicks from delivered impressions and measure effectiveness of search-side ad copy
Transaction conversion rate (TCR)	Number of clicks / Number of sales	Gauge the efficiency of generating sales from delivered clicks and measure effectiveness of website-side copy and design
Net revenue	Revenue – Cost	Compute the overall publisher contribution to a campaign
Return on ad dollars spent (ROA)	Net revenue / Cost	Compare the efficiency of generating net revenue (revenue less click charges) among publishers and campaigns
Average revenue per booking	Revenue / Number of bookings	Measure differences in revenue capture (booking = passenger in the case of Air France) and potentially identify different segments captured
Booking probability (take rate)	CTR × TCR	Gauge the probability of delivering a sale from delivered clicks

APPROACH

Optimizing an SEM campaign requires the following general steps:

1. *Summarize Metrics.* Summarize campaign metrics that will guide analysis and enable comparison of results.
2. *Optimize Publisher Portfolio.* Analyze the summarized campaign metrics to determine the best strategy to pursue for each publisher in the portfolio. There are four generic strategies:
 - a. Increase publisher funding by purchasing additional keywords.
 - b. Decrease publisher funding or eliminate from portfolio.
 - c. Improve search or website copy.
 - d. Adjust campaign tactics.
3. *Optimize Campaigns.* Analyze campaigns within each publisher to determine what campaign changes are needed to improve results.
4. *Calculate Impact.* Calculate the impact of campaign changes on metrics such as net revenue and ROA.
5. *Change and Measure.* Due to almost immediate access to data and ability to halt/modify campaigns on the fly, recommendation testing is not risky.

SUMMARIZE METRICS AND OPTIMIZE PUBLISHER PORTFOLIO

Suggested discussion questions:

- Which metrics will enable you to compare different publishers?
- Which are useful for comparing campaigns?
- What does each metric not measure?
- What do Air France's metrics suggest should be done to optimize its publisher portfolio?

The first step in optimizing a marketing campaign is to determine the metrics that will guide the analysis and enable comparison of SEM publishers and campaigns.

Students can be asked which metrics will help compare different publishers. CPC compares how much it costs to generate one user click from an ad on the publisher's site. However, clicks and cost are not the only important things to measure; results also need to be considered. Net revenue (Revenue – Cost) measures the overall financial contribution from a publisher, but it does not reveal whether the revenue was generated from many small sales or a few large ones.

For that we can turn to average revenue per booking (Revenue / Number of bookings), which may reveal that differing customer segments are captured by different publishers. Another measure is return on ad dollars spent (ROA) (Net revenue / Cost), which measures efficiency of generating net revenue—not profit—from each dollar of advertising.

The instructor can also guide students through a discussion of metrics to compare campaigns. CTR measures the efficiency of generating clicks from delivered impressions. Asking students if higher click-through rate is always better will provide opportunities to discuss the cost of additional clicks and the benefits of attracting the right sort of clicks to avoid paying for users who are unlikely to purchase.

TCR, which is calculated as Clicks / Number of sales, measures the efficiency of generating a sale from each click. CTR and TCR can be multiplied to generate a probability of booking, which shows the likelihood that each click will end in a sale.

Pivot tables can be used to summarize these metrics for each publisher as seen in **Table TN3**.

Table TN3: Summarized Metrics by Publisher

Publisher Name	Sum of Net Revenue	Average Cost Per Click	Total Volume of Bookings	Average Revenue Per Booking	ROA	Probability of Booking	Sum of Click Charges	Cost/Booking
Google - Global	\$808,603	\$1.66	797	\$1,166	669%	0.0441%	\$120,947	\$151.75
Google - US	\$1,391,841	\$1.84	1,550	\$1,126	394%	0.0402%	\$353,641	\$228.16
MSN - Global	\$133,364	\$1.08	129	\$1,128	1097%	0.0922%	\$12,160	\$94.27
MSN - US	\$165,451	\$1.49	140	\$1,297	1028%	0.0823%	\$16,098	\$114.99
Overture - Global	\$365,789	\$1.06	372	\$1,156	569%	0.0021%	\$64,296	\$172.84
Overture - US	\$205,457	\$1.19	289	\$1,202	145%	0.0017%	\$141,976	\$491.27
Yahoo - US	\$836,091	\$1.01	662	\$1,333	1810%	0.0709%	\$46,198	\$69.79
Grand Total	\$3,906,597	\$1.47	3,939	\$1,184	517%	0.0094%	\$755,316	\$191.75

If no publisher metrics were available, a reasonable approach might be to allocate the marketing budget based on the search engines' market share. Once data is available, however, it can be reviewed to identify standouts for cost and probability of booking.

Ask students what Air France's metrics suggest about optimizing the advertising portfolio. Three key observations should be discussed:

- Google-US had high net revenue and volume, but it is also the most expensive.
- Yahoo had the highest ROA and high net revenue.
- MSN outperformed Overture mainly due to differences in probability of booking.

One way to use the summarized metrics to optimize publishers is to create a bubble chart showing the CPC, probability of booking, and total costs for each. Probability of booking will be the X (or horizontal) axis and CPC will be the Y (or vertical) axis. The bubble size will represent total costs (or total funding). Higher-cost publishers will be near the top, and those with the highest probability of booking will be to the right.

After all the publishers are plotted, they can be divided into quadrants, which will help determine the appropriate strategy. A simpler approach is to continue with only the highest ROA keywords, campaigns, and publishers, but a more sophisticated approach—asking why one is better than another and applying the appropriate strategy—will yield better results.

Create four quadrants by drawing a horizontal line at the publishers' average CPC and a vertical line at the average probability of booking. The quadrant in which a publisher falls will suggest a generic strategy according to **Table TN4**.

Table TN4: Publisher Strategy by Quadrant

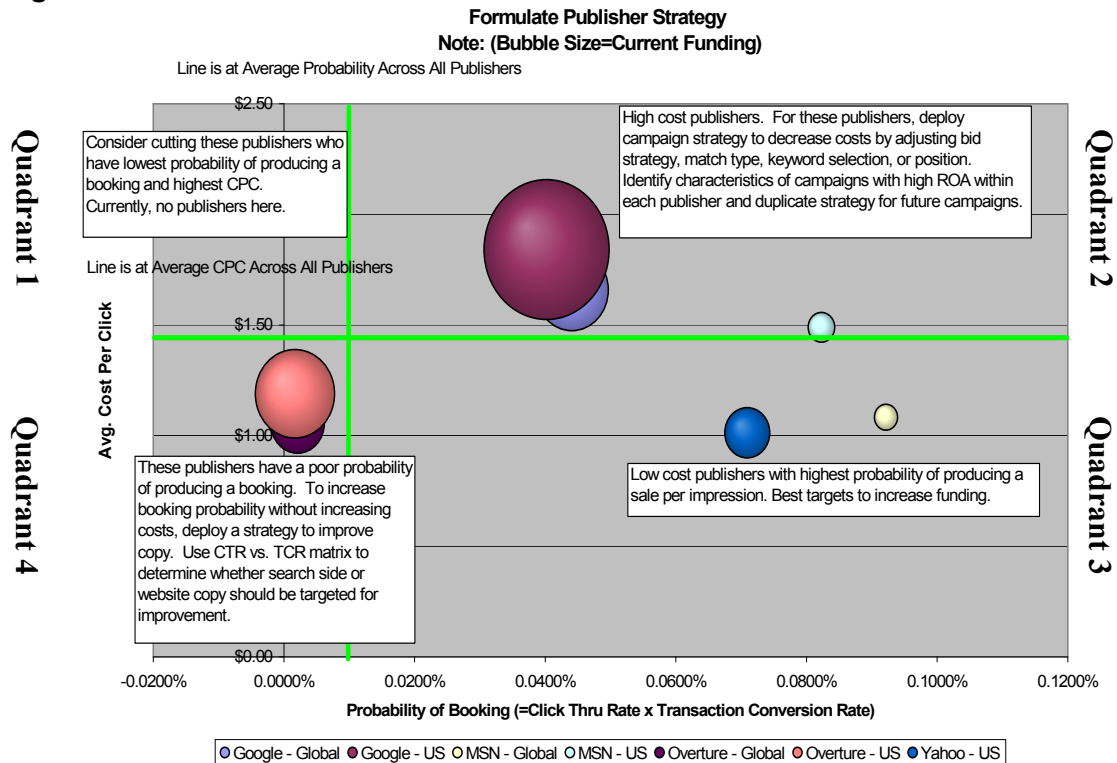
Quadrant	Probability of Booking	CPC	Publisher Type	Publisher Strategy
1	Low	High	Poor at Generating Sales and High Cost	Fund Less or Cut
2	High	High	Excellent at Generating Sales, but High Cost	Adjust Campaign Tactics
3	High	Low	Low Cost, Excellent at Converting Impression to Sale	Fund More
4	Low	Low	Low Cost, but Poor at Generating Sales	Improve Search of Site Side Copy

As shown in **Figure TN9**, CPC and probability to produce a sale differ significantly among the publishers in Air France's portfolio. There are no publishers in Quadrant 1 (low probability of booking and high cost). In Quadrant 2 (high probability of booking and high cost), Google-Global, Google-US, and MSN-US produce sales but can be optimized by making tactical campaign changes to lower their high cost.

Yahoo-US and MSN-Global are in Quadrant 3 (high probability of booking and low cost). These are attractive publishers, but the small bubble size means they are not getting much budget. Griffin's first recommendation should be to fund these publishers at a higher level. Students can be asked if it would make sense to move everything to Yahoo or MSN-Global since they are performing very well. There would be two potential problems with that approach: those two publishers may not address segments the others can, and there may not be enough inventory (or at least enough high-quality inventory) to use the entire marketing budget.

Finally, Quadrant 4 (low probability of booking and low cost) publishers, Overture-US and Overture-Global, require changes to improve CTR and/or TCR through copy improvements.

Figure TN9: Publisher Bubble Chart



OPTIMIZING CAMPAIGNS—OVERTURE

Suggested discussion questions:

- What is the problem with Overture campaigns?
- What actions should be taken with Overture campaigns?
- Should all low-CTR and low-TCR campaigns be eliminated?

As a publisher, Overture is low cost but has a lower take rate than others, which puts it in Quadrant 4. The instructor may ask students how they would isolate the reason for the lack of effectiveness. Before trying to optimize Overture campaigns, it is important to disaggregate the take rate into CTR and TCR so the areas of difficulty can be identified. Start by creating a bubble chart of Overture campaigns like the one created for publishers (X axis = CTR, Y axis = TCR, Bubble size = Net revenue). Divide the chart into four quadrants using the average CTR for all publishers as the vertical line and average TCR for all publishers as the horizontal line (see **Figure TN10**).

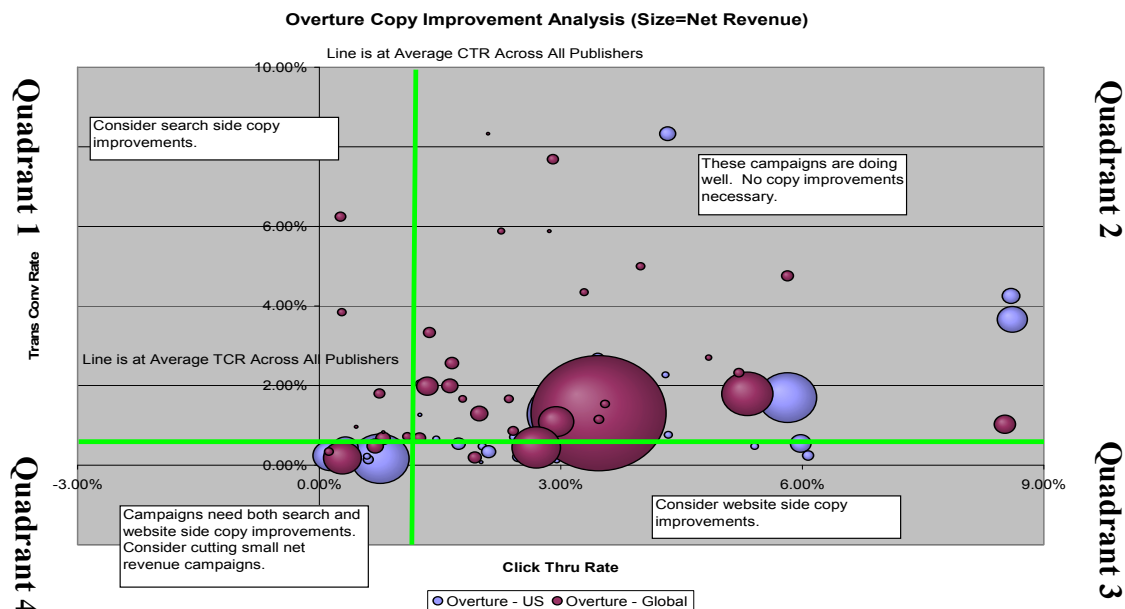
Campaigns in Quadrant 1 have a high transaction conversion rate, which means that once prospects reach the website there is an above-average rate of conversion to customers; however, the click-through rate is below average, so changes to the search-side copy should be tested.

Quadrant 2 campaigns are above average on both CTR and TCR, so there is less urgency to make major changes.

Campaigns located in Quadrant 3 are getting users to click on the ad, but too few are purchasing tickets. Air France should first consider improvements to the website to encourage greater conversion, but search-side copy should also be reviewed to confirm that it is attracting only targeted users.

Quadrant 4 campaigns are underperforming on both CTR and TCR. Those that generate little net revenue (i.e., small bubble size) may be eliminated. Campaigns that generate more net revenue should be reviewed for opportunities to improve the copy of the search ad and the website.

Figure TN10: Overture Campaign Bubble Chart



In summary, the first recommendation is to increase the take rate for Overture-US and Overture-Global campaigns.

OPTIMIZING CAMPAIGNS—GOOGLE AND MSN-US

Suggested discussion questions:

- What is the problem with Google and MSN-US campaigns?
- What metrics will be useful in optimizing these campaigns?
- How should the problems be addressed?

Overall Google and MSN-US were effective at producing bookings, but they were high cost. Understanding the common tactics of high-performing campaigns would enable Air France to apply them to low-performing campaigns within the same publisher and improve its overall ROA. The first step in optimizing their campaigns is to create a pivot table to summarize the tactics of campaigns with a high ROA that are performing well despite the high CPC of these publishers (see **Table TN5**). Tactics include keyword, match type, and bid strategy.

Table TN5: Google-US Tactics of Top 10 Campaigns**Average of Return on Ad Dollar Spent**

Campaign	Keyword	Match Type	Avg. Pos.	Bid Strategy	ROA
Geotargeted San Francisco	paris cheap airline	Broad	1.00	Position 5–10	32,237%
Air France Branded	air france us	Broad	1.02	Position 5–10	23,081%
Geotargeted New York	france airline ticket	Broad	1.54	Position 5–10	18,307%
Geotargeted Miami	france airfare sale	Broad	1.00	Position 5–10	16,915%
Geotargeted DC	france flights	Broad	1.20	Position 5–10	15,214%
Geotargeted Detroit	international airfares	Broad	3.01	Position 5–10	13,899%
Geotargeted Boston	paris cheap ticket	Broad	1.83	Position 5–10	13,070%
Geotargeted Houston	paris cheap flights	Broad	1.32	Position 5–10	12,895%
Google Yearlong 2006	rabat flights	Broad	1.14	Position 1–4	9,864%
Geotargeted Philadelphia	paris flight	Broad	1.71	Position 5–10	8,787%
Geotargeted Chicago	paris ticket	Broad	1.26	Position 5–10	6,290%
Geotargeted Seattle	paris tickets	Broad	1.69	Position 2–5	4,124%
Geotargeted Los Angeles	france air flight	Broad	1.14	Position 5–10	3,793%
French Destinations	air france to nice	Broad	1.08	Position 2–5	2,312%
Paris & France Terms	air france tickets paris	Broad	1.01	Position 2–5	1,467%

Ask students to identify the tactics of the high-ROA campaigns at Google-US. The following points should be discussed:

- They were geotargeted or Air France–branded.
- Their keywords included “france” or “paris.”
- They used broad match types.
- They used a bid strategy of Positions 5–10, which resulted in an average position of 1.40.

The low bid strategy and high resulting position was likely due to Google’s preference for high-CTR ads. Although it bid only for Positions 5–10, Air France likely had higher CTR for keyword combinations involving “paris,” “france,” or “air france” and thus were positioned higher than indicated by their bid strategy.

Applying the same approach to Google-Global campaigns yields **Table TN6**. The campaigns with the highest ROA had keyword combinations that included “air france” or Air France flight destinations, used broad match types, and employed a bid strategy of Positions 1–3. Testing these tactics on lower-ROA campaigns would be a reasonable way to improve cost effectiveness.

Table TN6: Google-Global Tactics of Top 10 Campaigns

Average of Return on Ad Dollar Spent

Campaign	Keyword	Match Type	Avg. Pos.	Bid Strategy	ROA
Air France Global Campaign	[airfrance]	Exact	2.00	Position 1–3	71,906%
	air france airlines	Broad	1.42	Position 1–3	12,326%
	lyon airfare	Broad	1.31	Position 1–3	12,120%
	cont:air france global campaign::france	N/A	4.84		11,684%
	nice flight	Broad	1.55	Position 1–3	10,728%
Air France Brand & French Destinations	air france site	Broad	1.03		28,410%
	air france com	Broad	1.11		5,605%
	airfrance	Broad	1.02		2,579%
	airfrance com	Broad	1.12		2,128%
	trip to paris france	Broad	1.31		2,124%

MSN-US campaigns can also be analyzed using the same pivot table (see **Table TN7**). The highest-return campaigns on MSN-US used “air france” as a keyword along with broad match types. Using these tactics with other MSN-US campaigns should help lower their costs and improve their returns.

Table TN7: MSN-US Tactics of Top 10 Campaigns

Average of Return on Ad Dollar Spent

Campaign	Keyword	Match Type	Avg. Pos.	Bid Strategy	ROA
Air France Brand & French Destinations	air travel in france	Broad	6.08		6,678%
	airfrance com	Broad	1.15		3,059%
	air france airlines	Broad	1.63		2,735%
	paris airfare	Broad	3.44		1,948%
	airfrance	Broad	1.19		1,703%
	air france	Broad	1.30		1,235%
	air france com	Broad	1.40		1,201%
	air france airline	Broad	1.13		1,081%
	french airlines	Broad	1.01		833%

Griffin’s second recommendation should be to lower the cost and improve the ROA of campaigns for Google-US, Google-Global, and MSN-US by applying tactics of successful campaigns to less successful ones.

KAYAK

Suggested discussion questions:

- How does Kayak's performance compare to the other publishers?
- Would you recommend more or less funding for the Kayak SEM campaign?

Kayak search and clickstream data was not integrated with DoubleClick, so the aggregated data for one week was all that was available. Kayak's CTR of 8%, given in the case, is essential to calculating the figures in **Table TN8**.

Table TN8: Kayak Summary

Time period is one week						
Search Engine	Clicks	Media Cost	Total Bookings	Avg Ticket	Total Revenue	Net Revenue
Kayak	2,839	\$3,567.13	208	\$1,123.53	\$233,694.00	\$230,126.87

Note: Kayak data is not available through DoubleClick. Only one week of data (above) is available and does not include campaign specifics. Due to differences in metrics and analysis period, Kayak data is analyzed separately from other publishers.

Kayak Trans. Conv. Rate	7.33%
Average Publisher TCR	0.77%
Kayak CPC	\$1.26
Average Publisher CPC	\$1.47

Kayak's delivered impressions are unknown, so the CTR cannot be determined. However, with a TCR 10x the average publisher's,

Recommendation 4 is to increase marketing through Kayak.

CALCULATING IMPACT

Suggested discussion questions:

- How can you estimate the impact of changing Air France's Internet marketing campaigns?

Using the data in the Excel file, we can calculate the expected impact of the changes in Air France's Internet marketing. This teaching note has reviewed four recommendations:

- *Recommendation 1:* Increase spending with MSN-Global and Yahoo-US, publishers that have a high take rate and low CPC.
- *Recommendation 2:* Increase take rate for Overture-US and Overture-Global campaigns.
- *Recommendation 3:* Lower cost and improve ROA of campaigns for Google-US, Google-Global, and MSN-US.
- *Recommendation 4:* Increase marketing through Kayak.

The expected impact of Recommendation 1 is shown in **Exhibit TN1**. The figures shown assume MSN-Global and Yahoo-US will each increase their share of the total spend by 2.5 percentage points (from 1.6% to 4.1% for MSN-Global, and from 6.1% to 8.6% for Yahoo-US) and each of the other publishers will see a one percentage point decrease. All other things equal, Recommendation 1 will increase total net revenue over \$300,000 during the next year.

Exhibit TN2 shows the impact of Recommendation 2, increasing the take rate for Overture-US and Overture-Global campaigns. Assuming a roughly 20% improvement in take rate (from 0.0021% to 0.0025% for Overture-Global and from 0.0017% to 0.0020% for Overture-US), net revenue will increase an additional \$142,479, for a total increase of \$454,055.

Continuing with Recommendation 3, the results of lowering the cost and improving the ROA of Google-US, Google-Global, and MSN-US campaigns is shown in **Exhibit TN3**. Assuming that applying the tactics of campaigns with high ROA to other campaigns reduces total click charges by 15%, an additional \$70,000 in net revenue will be generated.

The impact of Recommendation 4, increase marketing through Kayak, is shown in **Exhibit TN4**. The projected impact of a 10% increase in Kayak marketing is additional net revenue of \$1,196,660.

ATTRIBUTION

One of the benefits of online marketing is its measurability, but measurability can lead to problems as well. For example, if the return for all keywords can be calculated, why not just eliminate the low-return keywords and focus all marketing spending on the best resources? **Exhibit TN5** shows the distribution of keywords versus sales in a representative SEM campaign in which the first four keywords (to the left of the dashed line) account for approximately 50% of sales.

Eliminating the other keywords that produce little revenue would make sense if the entire sale could be attributed to the keyword associated with the search that produced it. However, research has shown this is a faulty premise; users proceed on a search journey that moves through different search engines and different keywords, generally progressing from non-branded keywords (e.g., Napa Valley winery tours) at the beginning of a search to branded keywords (e.g., Robert Mondavi winery tours) before purchase. In Exhibit TN5 the four keywords that produced the most sales were branded keywords and the rest were non-branded.

Once an online purchase is understood as a journey, the problems of attributing (or giving credit for) 100% of the sale to the final keyword become clear. For a given consumer, each keyword search is not an independent event—without the non-branded keywords to the right of the dashed line in Exhibit TN5 the consumers would never have reached the branded keywords that finally closed the sale.

Attribution modeling is a new technology that enables online advertisers to untangle the search journey to ensure that sufficient investment is made in the “assist clicks” early in the process that lead consumers to complete a purchase.

Epilogue

Air France made the changes recommended in Exhibits TN1 through TN4. As a result of these changes:

- CPC decreased 19%.
- CTR increased 112%.
- Bookings increased without increasing ad dollars spent.

Summary Comments

The following five steps will help optimize Internet marketing campaigns:

1. Summarize campaign metrics.
2. Optimize publisher portfolio.
3. Optimize campaigns.
4. Calculate impact.
5. Act and measure.

The following guidelines will help improve overall revenue and ROA:

- Increasing funding with a publisher that has high probability of booking/low CPC will have greatest impact on net revenue and ROA.
- Low booking probability/low CPC publishers should be targeted for copy improvements.
- Copy improvements that increase CTR and/or TCR will have the next greatest impact on net revenue and ROA.
- High booking probability/high CPC publishers should be targeted with tactical campaign changes indicative of high ROA campaigns.

Exhibit TN1: Impact of Recommendation 1 (Increase Marketing with MSN-Global and Yahoo-US)

Recommendation 1: Increase Marketing with MSN Global and Yahoo US.

These publishers are characterized by a high probability to book and low CPC charges.

Assuming funding mix is adjusted by +2.5% for MSN Global and Yahoo US by purchasing additional keywords. For all other publishers the funding mix will be decreased by 1%.

=Improved Metric

Publisher Name	Click Charges Current Year	Current Year Mix %	Next Year Mix %	Click Charges Next Year	Current Year Impressions	Cost Per Impression Current year	Impressions Purchased Next Year	Probability of Booking	Bookings	Average Revenue Per Booking	Net Revenue
Google - Global	\$120,947	16.0%	15.0%	\$113,394	1,808,326	\$0.07	1,695,395	0.0441%	747	\$1,166	\$758,106
Google - US	\$353,641	46.8%	45.8%	\$346,087	3,855,689	\$0.09	3,773,338	0.0402%	1517	\$1,126	\$1,362,114
MSN - Global	\$12,160	1.6%	4.1%	\$31,043	139,979	\$0.09	357,342	0.0922%	329	\$1,128	\$340,454
MSN - US	\$16,098	2.1%	1.1%	\$8,545	170,120	\$0.09	90,302	0.0823%	74	\$1,297	\$87,824
Overture - Global	\$64,296	8.5%	7.5%	\$56,743	17,898,727	\$0.00	15,796,073	0.0021%	328	\$1,156	\$322,818
Overture - US	\$141,976	18.8%	17.8%	\$134,423	17,062,488	\$0.01	16,154,760	0.0017%	274	\$1,202	\$194,527
Yahoo - US	\$46,198	6.1%	8.6%	\$65,081	933,345	\$0.05	1,314,840	0.0709%	933	\$1,333	\$1,177,835
Grand Total	\$755,316	100.0%	100.0%	\$755,316	41,868,674	\$0.02	39,182,051	0.0476%	4202	\$1,184	\$4,218,172

Net Revenue Current Year	\$3,906,597
ROA Current Year	517%
Net Revenue Subsequent Year	\$4,218,172
ROA Current Subsequent Year	558%
Impact of Recommendation 1 on Net Revenue	\$311,576
Impact of Recommendation 1 on ROA	41%

Exhibit TN2: Impact of Recommendation 2 (Increase Take Rate for Overture-US and Overture-Global Campaigns)

Recommendation 2: Improve Probability of Booking for Overture US and Overture Global using search and site side copy changes.
 These publishers are characterized by a low CPC rate and low probability to book.

Assuming campaigns indentified in CTR and TCR matrix are improved as indicated, resulting in probability improvement of 20%.

=Improved Metric

Publisher Name	Click Charges Current Year	Current Year Mix %	Next Year Mix %	Click Charges Next Year	Current Year Impressions	Cost Per Impression Current year	Impressions Purchased Next Year	Probability of Booking	Bookings	Average Revenue Per Booking	Net Revenue
Google - Global	\$120,947	16.0%	15.0%	\$113,394	1,808,326	\$0.07	1,695,395	0.0441%	747	\$1,166	\$758,106
Google - US	\$353,641	46.8%	45.8%	\$346,087	3,855,689	\$0.09	3,773,338	0.0402%	1517	\$1,126	\$1,362,114
MSN - Global	\$12,160	1.6%	4.1%	\$31,043	139,979	\$0.09	357,342	0.0922%	329	\$1,128	\$340,454
MSN - US	\$16,098	2.1%	1.1%	\$8,545	170,120	\$0.09	90,302	0.0823%	74	\$1,297	\$87,824
Overture - Global	\$64,296	8.5%	7.5%	\$56,743	17,898,727	\$0.00	15,796,073	0.0025%	394	\$1,156	\$398,730
Overture - US	\$141,976	18.8%	17.8%	\$134,423	17,062,488	\$0.01	16,154,760	0.0020%	328	\$1,202	\$260,317
Yahoo - US	\$46,198	6.1%	8.6%	\$65,081	933,345	\$0.05	1,314,840	0.0709%	933	\$1,333	\$1,177,835
Grand Total	\$755,316	100.0%	100.0%	\$755,316	41,868,674	\$0.02	39,182,051	0.0477%	4323	\$1,184	\$4,360,651

Net Revenue Current Year	\$3,906,597
ROA Current Year	517%
Net Revenue Subsequent Year	\$4,360,651
ROA Current Subsequent Year	577%
Impact of Recommendation 1 and 2 on Net Revenue	\$454,055
Impact of Recommendation 1 and 2 on ROA	60%

Exhibit TN3: Impact of Recommendation 3 (Lower Cost and Improve ROA of Campaigns for Google-US, Google-Global, and MSN-US)

Recommendation 3: Improve ROA of campaigns for high CPC publishers (Google US, Google Global, and MSN US) by decreasing CPC rates. These publishers are characterized by a high probability to book and high CPC charges.

Assuming tactics of campaigns with high ROA from PivotTable are applied to other campaigns and click charges end up 15% less than anticipated

=Improved Metric

Publisher Name	Click Charges Current Year	Current Year Mix %	Next Year Mix %	Click Charges Next Year	Current Year Impressions	Cost Per Impression Current year	Impressions Purchased Next Year	Probability of Booking	Bookings	Average Revenue Per Booking	Net Revenue
Google - Global	\$120,947	16.0%	15.0%	\$96,385	1,808,326	\$0.05	1,695,395	0.0441%	747	\$1,166	\$775,115
Google - US	\$353,641	46.8%	45.8%	\$294,174	3,855,689	\$0.07	3,773,338	0.0402%	1517	\$1,126	\$1,414,027
MSN - Global	\$12,160	1.6%	4.1%	\$30,968	139,979	\$0.07	357,342	0.0922%	329	\$1,128	\$340,530
MSN - US	\$16,098	2.1%	1.1%	\$7,264	170,120	\$0.08	90,302	0.0823%	74	\$1,297	\$89,106
Overture - Global	\$64,296	8.5%	7.5%	\$56,743	17,898,727	\$0.00	15,796,073	0.0025%	394	\$1,156	\$398,730
Overture - US	\$141,976	18.8%	17.8%	\$134,423	17,062,488	\$0.01	16,154,760	0.0020%	328	\$1,202	\$260,317
Yahoo - US	\$46,198	6.1%	8.6%	\$64,957	933,345	\$0.04	1,314,840	0.0709%	933	\$1,333	\$1,177,959
Grand Total	\$755,316	100.0%	100.0%	\$684,913	41,868,674	\$0.02	39,182,051	0.0477%	4323	\$1,184	\$4,431,054

Net Revenue Current Year	\$3,906,597
ROA Current Year	517%
Net Revenue Subsequent Year	\$4,431,054
ROA Current Subsequent Year	587%
Impact of Recommendation 1,2,3 on Net Revenue	\$524,457
Impact of Recommendation 1,2,3 on ROA	69%

Exhibit TN4: Impact of Recommendation 4 (Increase Marketing with Kayak)

Recommendation 4: Increase Marketing with Kayak.

Publisher is characterized by a high probability to book and low CPC charges.

Assuming funding for Kayak is increased 10%.

=Improved Metric

Publisher Name	Estimated Annual Media Cost	Click Charges Next Year	Clicks Purchased Next Year	TCR	Bookings	Average Revenue Per Booking	Net Revenue
Kayak	\$185,491	\$204,040	162,391	7.33%	11,898	\$1,124	\$13,163,257

Net Revenue Current Year	\$11,966,597
ROA Current Year	6451%
Net Revenue Subsequent Year	\$13,163,257
ROA Current Subsequent Year	7096%
Impact of Recommendation on Net Revenue	\$1,196,660
Impact of Recommendation on ROA	645%

Exhibit TN5: SEM Keywords and Sales